

Techno-Economic Assessment for an Off-Grid Solar PV and Energy Storage System

For a 100,000-Person Refugee Camp Near Damascus, Syria

Prepared for: The United Nations High Commissioner for Refugees (UNHCR)

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Executive Summary

This report presents a techno-economic assessment for a proposed off-grid solar photovoltaic (PV) and battery energy storage system (BESS) to power a refugee camp for 100,000 individuals near Damascus. The analysis concludes that a **hybrid solar-PV and BESS are not only technically feasible but also significantly more economically favorable than a conventional diesel generator system over a 20-year project lifespan.**

Key Findings:

- **System Sizing:** The recommended system consists of an 11 **MWp solar PV array** coupled with a **47 MWh Li-ion battery storage system**.
- **Load Analysis:** The camp's peak demand is estimated at 4.35 MW, with a total daily energy consumption of approximately 54 MWh.
- **Economic Comparison:**
 - **Solar + BESS Scenario:** Net Present Cost (NPC) = **\$18.2 million**.
 - **Diesel Generator Scenario:** Net Present Cost (NPC) = **\$105 million**.
 - **LCOE:** Solar + BESS = \$0.13/kWh
 - **LCOE:** Diesel Generator = \$0.47/kWh
- **Savings:** The solar-PV-BESS system offers a present net cost saving of **approximately \$86.8 million**, representing a 83% reduction compared to the diesel-only scenario.
- **Recommendation:** It is strongly recommended that the UNHCR and the Government of Syria proceed with the implementation of the proposed solar-PV and battery storage system. This solution provides lower long-term costs, enhanced energy security, and significant environmental and social benefits.

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1. Introduction & Project Background

The ongoing influx of returning Syrian refugees necessitates the establishment of resilient and sustainable infrastructure. A stable and cost-effective power supply is critical for lighting, health services, water pumping, education, and overall safety within a refugee camp. This report assesses the feasibility of an off-grid solar PV system with battery storage as the primary power source, comparing it against conventional diesel generation.

2. Design Assumptions & Methodology

To conduct this analysis, several key assumptions were made based on industry standards and regional data.

2.1. Key Assumptions:

- **Camp Population:** 100,000 refugees.
- **Household Size:** Assumed at 5 people per prefab unit, resulting in **20,000 family units**.
- **Solar Resource:** Damascus has excellent solar insolation. We assume a conservative average of **5.5 peak sun hours (PSH) per day** year-round [Source: Global Solar Atlas, World Bank Group].
- **System Losses:** Inverter and system losses are estimated at 15%.
- **Days of Autonomy:** The battery system is designed to provide **2 full days of autonomy** to ensure power availability during prolonged cloudy periods.

- **Generator and solar PV occupies sample size of land.**
- **Diesel Generator Assumptions:**
 - **Fuel Cost:** \$1.20 per liter (based on current regional prices and logistical challenges).
 - **Generator Efficiency:** 3.5 kWh per liter of diesel.
 - **Generator Capital Cost:** \$400 per kW [Source: Industry average quotes].
 - **Operation & Maintenance (O&M):** \$0.05 per kWh for diesel O&M [Source: HOMER Pro default data].
- **Project Lifespan:** 20 years.

Discount Rate: A 6% discount rate is applied to account for the cost of capital and inflation [Source: Standard for humanitarian infrastructure projects].

3. Load Assessment & Power Requirements

The total electrical load for the camp is calculated as follows:

3.1. Detailed Load Calculation: see attached excel file

Segment	Sub-load	Qty	Rated Power (W)	Hours/Day	Daily Energy (Wh/d)	Daily Energy (kWh/d)	TOTAL POWER kW
A. Camps	TV	1	50	6	300	6000	1000
	Fan	2	50	12	1200	24000	2000
	LED Lights	5	10	6	300	6000	1000
	Small Refrigerator	1	200	8	480	9600	4000
	Sum				2280	45600	
					Total number of units	20,000	
B. Clinic	Refrigerator	2	750	24	36000	36	1.5
	LED Lights	10	10	18	1800	1.8	0.1
	Medical Equipment	2	10,500	12	252000	252	21
	Others	1	500	6	3000	3	0.5
	Sum				292800	292.8	
C. Pumping Water Supply	Electric Motors	3	25,000	6	450000	450	75
	Control Panel	2	5,000	6	60000	60	10
	Sum				510000	510	
D. School Lighting	LED Lights	500	10	8	40000	40	5
	Others	50	500	6	150000	150	25
	Sum				190000	190	
E. Street Lighting	LED Lights	100	150	12	180000	180	15
	Sum				180000	180	
							8153.1
						46772.8	
					Total Daily energy + 15% contingency	53788.72	
					Total yearly energy demand	19632882.8	

3.2. Load Profile Summary:

- **Peak Demand:** To account for simultaneous operation and future expansion, a **15% contingency** is added. Therefore, the design peak load is $4,153 \text{ kW} * 1.15 \approx 4,775 \text{ kW}$. We will use **4.35 MW** as a conservative rounded figure for sizing.
- **Total Daily Energy Consumption:** **~54 MWh/day** (after adding contingency).

4. System Sizing and Design

4.1. Solar PV Array Sizing:

The PV array must generate the total daily energy demand, accounting for system losses.

- Daily Energy Required = 53,788.72 kWh
- Required Solar Generation per Day = $53,788.72 \text{ kWh} / (1 - 0.15) = 63,280.8 \text{ kWh}$
- Solar Array Size = Daily Energy Required / Peak Sun Hours = $63,280.8 \text{ kWh} / 5.5 \text{ h} \approx \mathbf{11,735.72073 \text{ kWp}}$

A **12 MWp** system is proposed.

- **4.2. Battery Energy Storage System (BESS) Sizing:**

The BESS must cover the energy needs when solar is not available.

- Energy for One Night = (Daily Energy) * (Percentage of use at night). We assume ~70% of energy is needed outside of solar hours.
- Nighttime Energy Requirement = $53,788.72 \text{ kWh} * 0.70 \approx 37,652.104 \text{ kWh}$
- With 2 days of autonomy: $37,652 \text{ kWh} * 2 = 75,304 \text{ kWh}$
- To prevent deep discharging and prolong battery life, we assume a maximum Depth of Discharge (DoD) of 80%.
- Usable Battery Capacity Required = $75,304 \text{ kWh} / 0.80 = \mathbf{94,130 \text{ kWh}}$

- **Proposed System Configuration:**

- **Solar PV Array:** 12 MWp
- **Battery Storage (Li-ion):** 94 MWh
- **Power Conversion System (Inverters):** 11 MW (sized for peak load)

5. Economic Analysis

The analysis uses the Net Present Cost (NPC) method to compare the lifetime costs of the two scenarios over 20 years.

5.1. Scenario 1: Solar-PV + BESS System (Hybrid)

Parameter	Value	Units	Notes
PV CAPEX (\$)	\$ 11,911,756.54	\$	Including PV BOS/EPC
Battery CAPEX (\$)	18826052	\$	
Power electronics CAPEX (\$)	1240581.906	\$	inverter CAPEX
Subtotal CAPEX (\$)	31978390.44	\$	
Total CAPEX (\$)	\$ 31,978,390.44	\$	
Annual energy served by PV-battery (kWh/yr)	19632882.8	kWh/yr	
Replacement	\$ 18,826,052.00	\$	battery replace after 10 years
Annual fixed O&M (\$/yr)	50000	\$/yr	
Discount rate	0.06		
Project life (yrs)	20	yr	
Battery replacement years	10	yr	
NPC CAPEX	\$ 17,856,566.20	\$	
NPC of O&M over life (\$)	573496.0609	\$	Annuity present value
NPC of battery replacement	\$ 10,512,369.11	\$	
NPC total (\$)	\$ 28,942,431.37	\$	
LCOE (\$/kWh)	\$ 0.13	\$/kWh	

5.2. Scenario 2: Diesel Generators Only (Baseline)

Parameter	Value	Units	Notes
Annual demand served by diesel (kWh/yr)	19632882.8	kWh/yr	
Specific fuel consumption (L/kWh)	0.3	L/kWh	
Fuel price (\$/L)	1.2	\$/L	
Annual fuel (L/yr)	5889864.84	L/yr	
Annual fuel cost (\$/yr)	7067837.808	\$/yr	
Diesel CAPEX (\$)	6522480	\$	cost of generator sets
Variable O&M (\$/kWh)	0.05	\$/kWh	Adjust if you used different cell
Annual variable O&M (\$/yr)	\$ 981,644.14	\$/yr	
Annual diesel cost (\$/yr)	\$ 8,049,481.95	\$/yr	Excludes CAPEX
Total Annual cost	\$ 9,031,126.09	\$/yr	
Discount rate	0.06		
Project life (yrs)	20	yr	
NPC CAPEX	\$ 2,033,740.10	\$	
NPC of O&M over life (\$)	\$ 103,586,304.74	\$	Annuity present value
NPC total (\$)	\$ 105,620,044.84	\$	
LCOE (\$/kWh)	\$ 0.47	\$/kWh	

Note: The NPC for diesel is extremely sensitive to the fuel price.

6. Recommendations

Based on technical and economic analysis, Energy Solutions Consulting Group strongly recommends proceeding **with the Hybrid Solar-PV and BESS**

Solution: The hybrid system offers a Net Present Cost saving of **\$87 million** over 20 years compared to a diesel-only

7. Conclusion

The proposed 12 MWp solar PV and 40 MWh battery storage system is both technically sound and economically superior for powering the 100,000-person refugee camp near Damascus. While the initial investment is higher than a diesel generator setup, the elimination of volatile and high fuel costs results in massive long-term savings. Furthermore, this solution provides silent, clean, and reliable power, improving the quality of life for refugees and aligning with global sustainable development principles. The hybrid system offers energy independence and resilience, which is of paramount importance in a humanitarian context.

References:

<https://www.irena.org/Costs>

<https://globalsolaratlas.info/>

<https://about.bnef.com/blog/energy-storage-system-costs-survey-2024/>

<https://www.homerenergy.com/>

<https://www.unocha.org/syria>

<https://ppp.worldbank.org/public-private-partnership/financing/issues/discount-rates>

<https://pvwatts.nrel.gov/>